• **IDLE—HIGH IDLE AFTER STEADY STATE DRIVING IN EXTREME COLD AMBIENT TEMPERATURES—VEHICLES EQUIPPED WITH 4.2L ENGINE BUILT THROUGH 1/24/2000** • **ENGINE—4.2L—PCV SYSTEM ICING—AFTER STEADY STATE DRIVING IN EXTREME COLD AMBIENT TEMPERATURES—VEHICLES EQUIPPED WITH 4.2L ENGINE BUILT THROUGH 1/24/2000**	**Article No. 01-20-1**

FORD: 1997-2000 F-150

ISSUE

Some of the vehicles produced through 11/13/1996 identified in this TSB article should have been serviced under recall 97S96 or 98R04. Vehicles produced after 11/13/1996 have production level coolant heated fittings in the Upper Intake. In those extreme cases where throttle body icing due to PCV backflow is not completely eliminated, a new designed heated PCV valve has been developed that will make these vehicles even less susceptible to PCV freezing and the possibility of throttle body icing.

ACTION

Verify that the condition is not caused by throttle body sludging, throttle cable binding, or interference between an improperly positioned floor mat and accelerator pedal. If no other conditions are identified, check for evidence of PCV system backflow by inspecting the inside of the clean air tube at the PCV closure hose connection for the presence of moisture, oil, or oily stains. If evidence of system backflow is found, recall 97S96/98R04 should be performed (if required) and a new production level heated PCV valve assembly should be installed. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

First determine which model year and which level of repair is needed.

SERVICE PROCEDURE # 1

1997 model and 1998 MODEL YEAR F150 built prior to 11/13/1996. (Vehicles that have NOT been modified by Recall 97S96 & 98R04)

Recall Kits F8PZ-6A603-BA (Kit BB) Coolant Heated Fitting & F7TZ-6A603-AA (Kit AA) Tube-In-Tube required.

1. Perform Recall 97S96/98R04 and NOTE the following changes to the instructions.
2. In Step 21, install the new heated “Horseshoe” style PCV valve assembly (YL3Z-6A666-AA) in the Valve cover. (TSB Figure 1).
3. In Step 28, connect the 1/4 inch coolant hose to the heater supply hose “T” fitting and route it to the heated PCV assembly in the valve cover. (TSB Figure 1) and 11-12 (Recall Instructions).

NOTE

THE 1/4 INCH HOSE CAN BE CONNECTED TO EITHER CONNECTION ON THE HEATED PCV ASSEMBLY, FLOW DIRECTION IN THE ’HORSESHOE’ FITTING DOES NOT MATTER.

4. Disregard Step 29. Instead, install a 1/4 x 22.5 inch (381175-S225A) coolant hose (not supplied in kit and cut to length as required) to the other fitting on the heated PCV valve assembly and route it to the heated PCV fitting on the Upper Intake manifold. Connect to either fitting; flow direction does not matter. Secure with hose clamps (E9ZZ-8287-AA) not supplied in kit. (See TSB Figure 1 for suggested routing.)
5. Install a 1/4 x 22.5 inch (381175-S225A-Kit) coolant hose to the remaining PCV fitting (upper intake) and route it back to the heater (return) hose “T” fitting. Secure with clamps (E9ZZ-8287-AA-Kit). Figures 11-12 (Recall Instructions).
6. Complete the rest of the steps in the recall procedure starting with Step 30.

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CAUTION
FAILURE TO FOLLOW THE CORRECT RE-FILL PROCESS WHEN TOPPING OFF THE COOLING SYSTEM COULD CAUSE AN ENGINE OVERHEAT CONDITION. THE INSTALLATION OF ADDITIONAL HOSES AND PARTS MAY TRAP AIR IN THE COOLING SYSTEM THAT SHOULD BE PURGED. FOLLOW THE INSTRUCTIONS LISTED IN THE APPROPRIATE MODEL YEAR WORKSHOP MANUAL, COOLING SECTION 303-03, COOLING SYSTEM DRAINING, FILLING AND BLEEDING PROCEDURE.

SERVICE PROCEDURE # 2

1997 model 1998 MODEL YEAR F150 built prior to 11/13/1996. (Vehicles that HAVE already been modified by recall 97S96 and 98R04). (Service Kit not required)

NOTE
THIS SERVICE PROCEDURE WILL ROUTE ENGINE COOLANT FROM THE HEATER SUPPLY HOSE TEE FITTING TO THE (NEW) HEATED PCV VALVE ASSEMBLY IN THE VALVE COVER, THEN TO THE HEATED PCV FITTING IN THE UPPER INTAKE MANIFOLD AND BACK TO THE HEATER RETURN HOSE TEE FITTING.

1. Install memory saver, then disconnect battery negative cable.
2. Drain enough coolant to allow for heater hose modification.
3. Remove the old PCV valve from the valve cover and replace it with the new "Horseshoe" type heated PCV valve assembly (YL3Z-6A666-AA). (TSB Figure 1).
4. Connect the PCV valve vacuum hose to the new PCV valve assembly. (TSB Figure 1).
5. Disconnect the 1/4 inch coolant hose (heater supply hose "T" splice) from the heated PCV fitting (upper intake manifold) and route it to the new "Horseshoe" type heated PCV valve assembly in the valve cover. Cut hose to length as required and make sure hose is routed so no kinks are present. Hose can be connected to either fitting, flow direction does not matter. Secure with constant tension clamps (E9ZZ-8287-AA). (See TSB Figure 1 for suggested Routing).
6. Connect a 1/4 x 22.5 inch (381175-S225A) coolant hose, cut to length as required, to the other fitting on the heated PCV valve assembly (valve cover) and route it to the open fitting on the heated PCV fitting in the Upper Intake Manifold. Secure with clamps (E9ZZ-8287-AA). (TSB Figure 1).
7. Connect battery negative cable, then remove memory saver.
8. Refill engine cooling system.

CAUTION
FAILURE TO FOLLOW THE CORRECT REFILL PROCESS WHEN TOPPING OFF THE COOLING SYSTEM COULD CAUSE AN ENGINE OVERHEAT CONDITION. THE INSTALLATION OF ADDITIONAL HOSES AND PARTS MAY TRAP AIR IN THE COOLING SYSTEM THAT SHOULD BE PURGED. FOLLOW THE INSTRUCTIONS LISTED IN THE APPROPRIATE MODEL YEAR WORKSHOP MANUAL, COOLING SECTION 303-03, COOLING SYSTEM DRAINING, FILLING AND BLEEDING PROCEDURE.

9. Start and run engine until normal operating temperature is reached.
10. Check all coolant hoses for leaks. Repair as necessary.
11. Check coolant level. Top-off as necessary.

SERVICE PROCEDURE # 3

1998 MODEL YEAR F150 built prior to 11/13/1996 through 2000 Model Year F150 (to end of engine production running change 1/24/00).

Service Kit F8PZ-6A603-BA (Kit BB) Coolant Heated Fitting required.

NOTE
THIS SERVICE PROCEDURE WILL ROUTE ENGINE COOLANT FROM THE NEW INSTALLED HEATER SUPPLY HOSE TEE FITTING TO THE (NEW) HEATED PCV VALVE ASSEMBLY IN THE VALVE COVER, THEN TO THE HEATED PCV FITTING IN THE UPPER INTAKE MANIFOLD AND BACK TO THE HEATER RETURN HOSE FITTING IN THE NEW INSTALLED HEATER WATER OUTLET TUBE ASSEMBLY.

1. Install memory saver, then disconnect battery negative cable.
2. Drain enough coolant to allow for heater hose modification.

3. Remove the old PCV valve from the valve cover and replace it with the new "Horseshoe" type PCV valve assembly (YL3Z-6A666-AA). (TSB Figure 1).
4. Connect the PCV valve vacuum hose to the new PCV valve assembly. (TSB Figure 1).
5. Locate the heater (supply) hose and mark hose 10 1/2 inches from the end of the hose at the dash panel. (See Figure 10 (Recall Instructions) and TSB Figure 2).
6. Cut heater (supply) hose at marked location and install one tee (supplied in Recall Kit). Position the 1/4 inch tap as shown. (TSB Figure 2 and 10-11 (Recall Instructions). Secure tee with clamps supplied in the Recall Kit.
7. Install a 1/4 x 22.5 inch (381175-S225A) coolant hose to the 1/4 inch tap on the tee and route it to the 1/4 inch fitting on the heated PCV valve assembly. Hose can be connected to either 1/4 inch connector, flow direction in the PCV assembly does not matter. Make sure hose is routed so no kinks are present. (See TSB Figure 1 for suggested Routing.) Secure with constant tension clamps (E9ZZ-8287-AA).
8. Disconnect both 1/4 inch coolant hoses from the heater water outlet tube assembly. Save the clamps for reinstall and leave the hoses connected at the other end for re-routing.
9. Remove the old heater water outlet tube assembly and install the new assembly (1L3Z-18663-BA) with only one 1/4 inch hose connection. (TSB Figure 3).
10. Re-route and install one of the 1/4 inch coolant hoses to the other fitting on the heated PCV valve assembly (valve cover) and install the other 1/4 inch hose to the new installed heater water outlet tube assembly. This completes the connections to the heated fittings in the Upper Intake Manifold. Secure with clamps. (TSB Figure 1).
11. Connect battery negative cable, then remove memory saver.
12. Refill engine cooling system.

CAUTION
FAILURE TO FOLLOW THE CORRECT REFILL PROCESS WHEN TOPPING OFF THE COOLING SYSTEM COULD CAUSE AN ENGINE OVERHEAT CONDITION. THE INSTALLATION OF ADDITIONAL HOSES AND PARTS MAY TRAP AIR IN THE COOLING SYSTEM THAT SHOULD BE PURGED. FOLLOW THE INSTRUCTIONS LISTED IN THE APPROPRIATE MODEL YEAR WORKSHOP MANUAL, COOLING SECTION 303-03, COOLING SYSTEM DRAINING, FILLING AND BLEEDING PROCEDURE.

13. Start and run engine until normal operating temperature is reached.
14. Check all coolant hoses for leaks. Repair as necessary.
15. Check coolant level. Top-off as necessary.

PART NUMBER	PART NAME
YL3Z-6A666-AA	"Horseshoe" Type Heated PCV Valve Assembly
F8PZ-6A603-BA	Recall Kit "BB" Service Kit
F7TZ-6A603-AA	Recall Kit "AA" Service Kit
1L3Z-18663-BA	Heater Water Outlet Tube Assembly
381175-S225A	1/4 x 22.5 inch Coolant Hose
E9ZZ-8287-AA	Constant Tension Clamp

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
012001A	Install New PCV Assembly, And Water Outlet Tube	0.9 Hr.
012001B	Install Recall Service Kit, And New PCV Assembly	2.1 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
6758	55

OASIS CODES: 499000, 608000, 608600, 619400

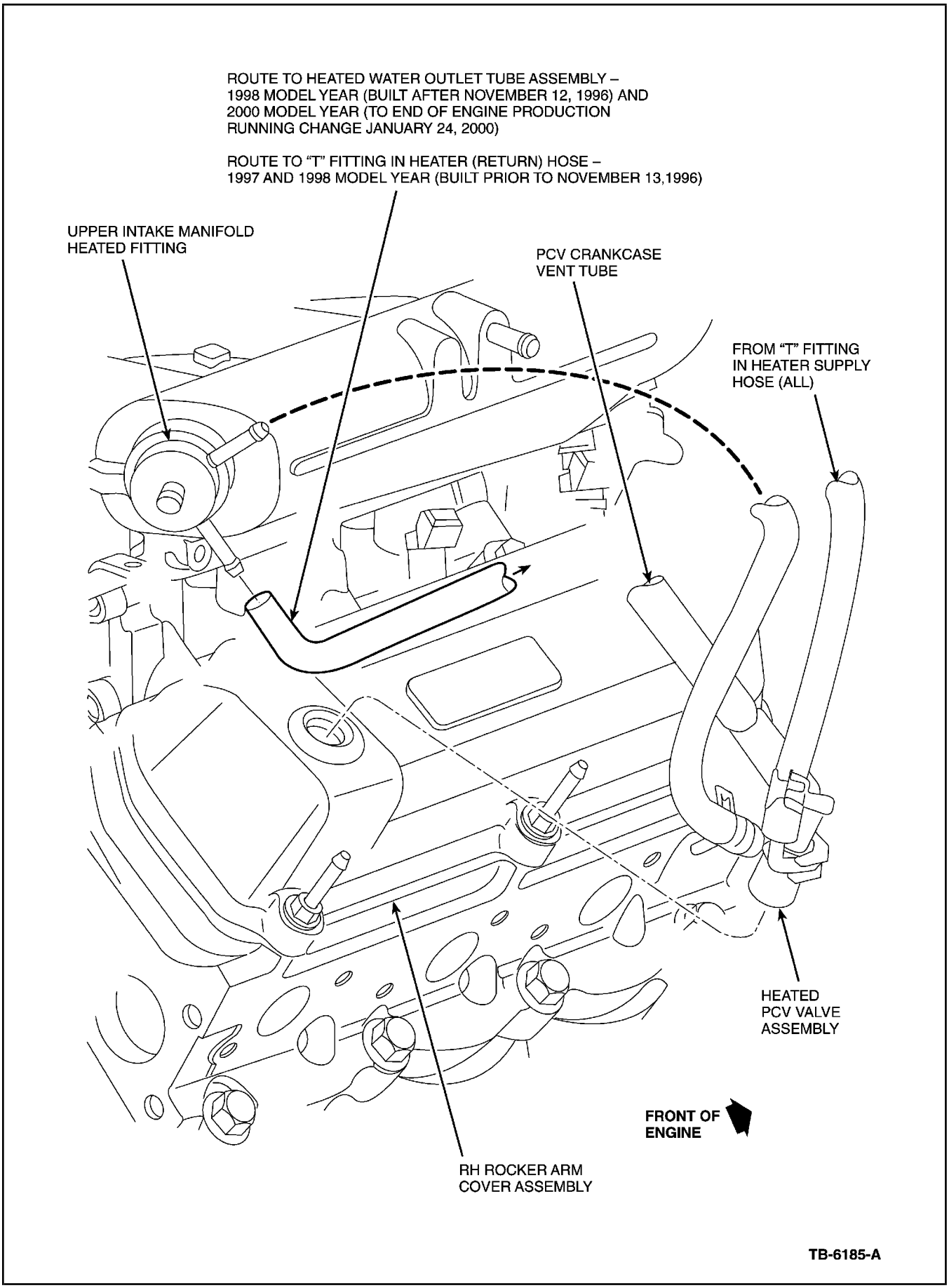


Figure 1 - Article 01-20-1

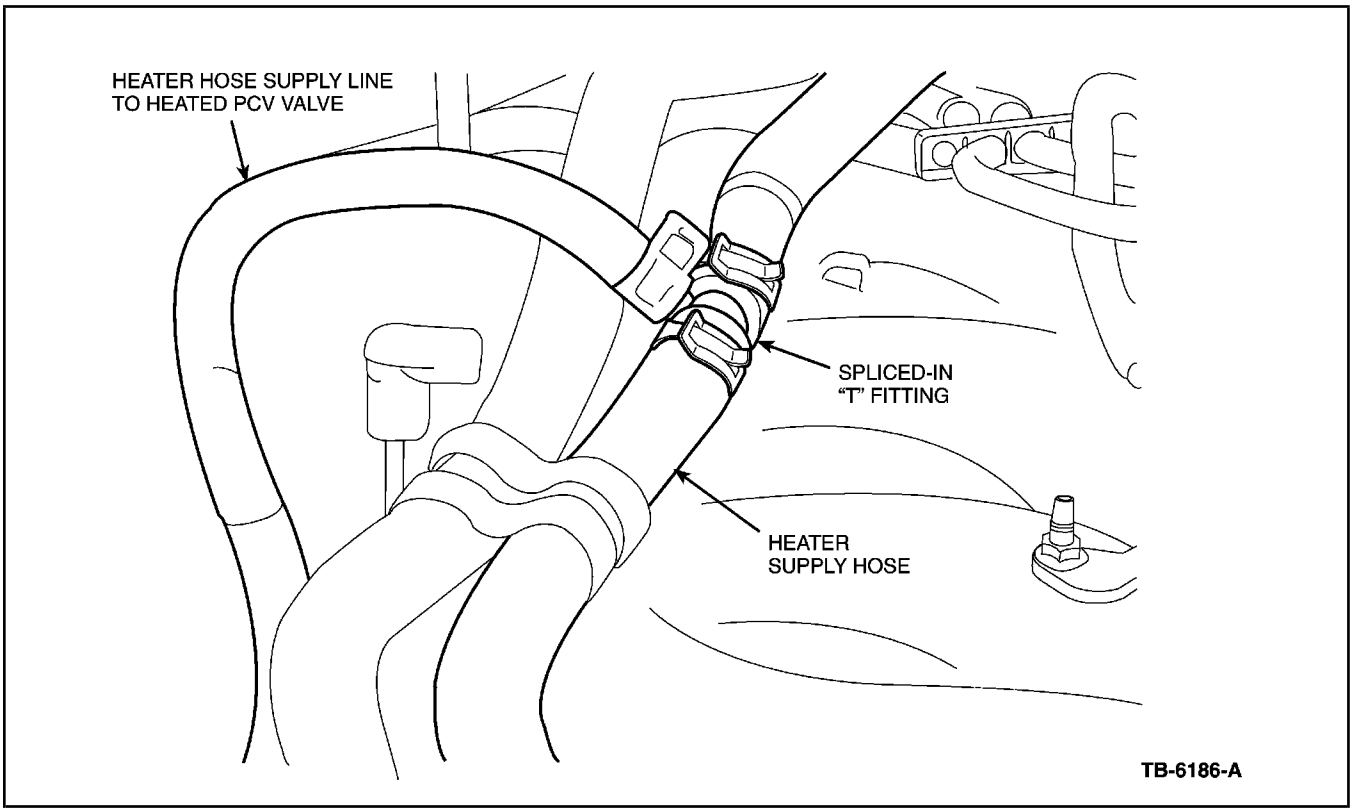


Figure 2 - Article 01-20-1

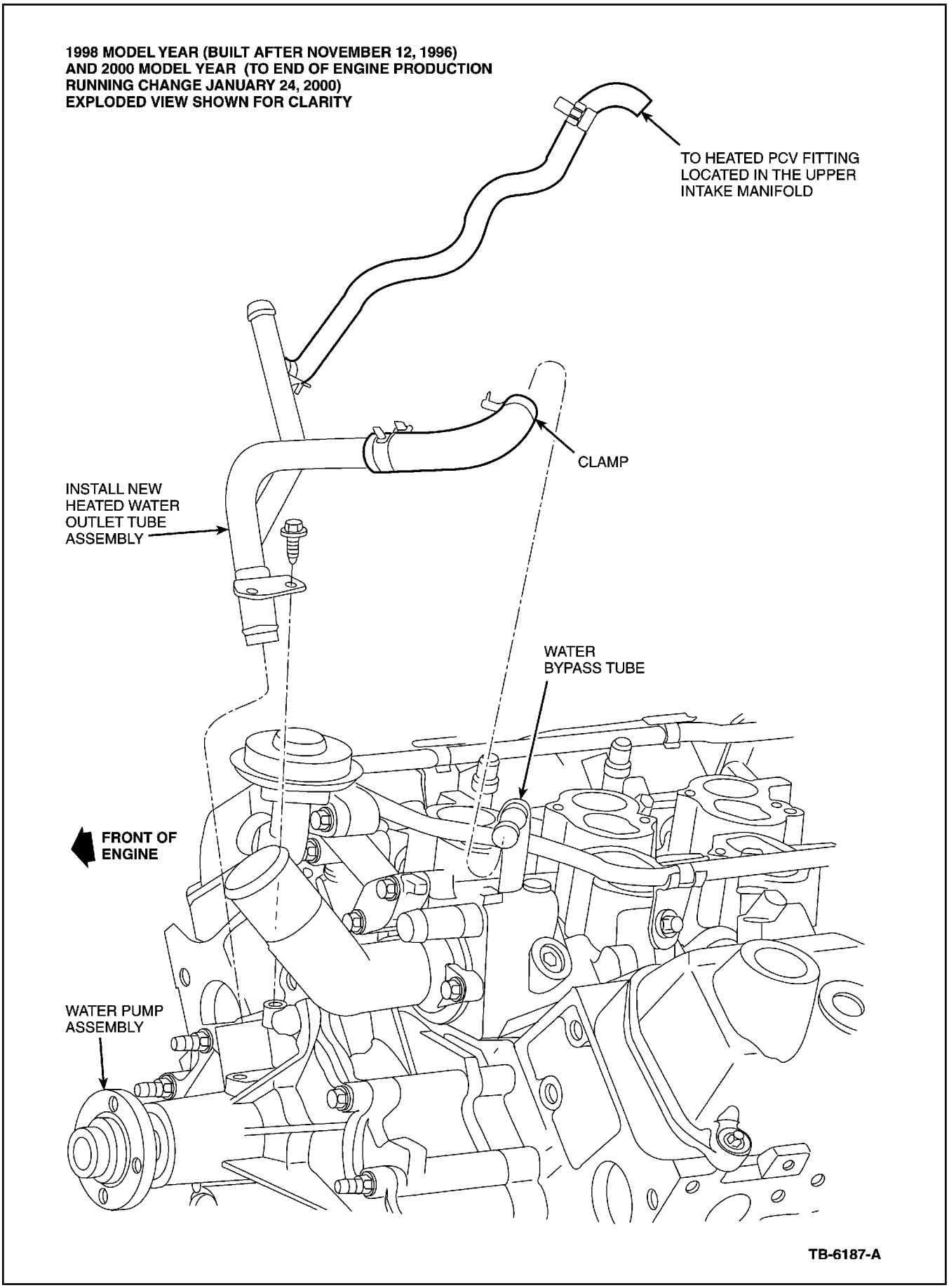


Figure 3 - Article 01-20-1